

Form Number : _____

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CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 - 2020)

PRE-MEDICAL : ENTHUSIAST COURSE

PHASE - MEA,B,C,D,F,G,P,Q,R,S,T,U & MEPS

Test Type : COMMON TEST(SRG)

Test Pattern : NEET(UG)

TEST DATE : 08 - 12 - 2019

Time : 3 Hours

Maximum Marks : 720

Important Instructions

Do not open this Test Booklet until you are asked to do so

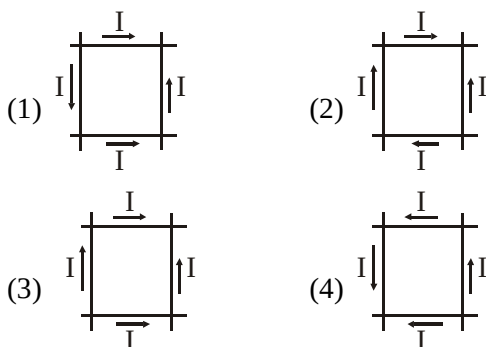
1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Student can not use log tables and calculators or any other material in the examination hall.
3. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
4. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
5. Each correct answer carries 4 marks, while one mark will be deducted for each incorrect answer.
6. Use the OMR provided separately for answering the question.

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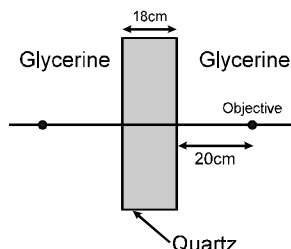
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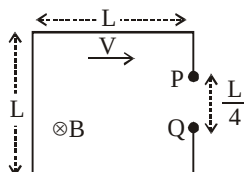
1. The figure shows four different sets of wires that cross each other without actually touching. The magnitude of the current is the same in all four cases, and the directions of current flow are as indicated. For which configuration will the magnetic field at the center of the square formed by the wires be equal to zero?



2. Given that, velocity of light in quartz = 1.5×10^8 m/s and velocity of light in glycerine = $(9/4) \times 10^8$ m/s. Now a slab made of quartz is placed in glycerine as shown. What is the shift produced by slab?



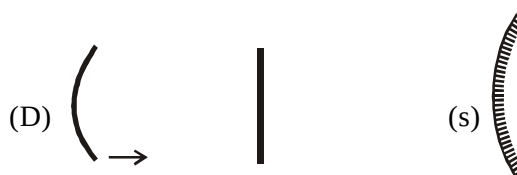
- (1) 6 cm (2) 3.55 cm (3) 9 cm (4) 2 cm
3. The loop shown moves with a constant velocity 'V' in a uniform magnetic field to magnitude 'B' directed into the paper. The potential difference between P and Q is :-



- (1) $e = \frac{3}{4}BLV$, Q is positive with respect to P
 (2) $e = \frac{1}{4}BLV$, P is positive with respect to Q
 (3) $e = 0$
 (4) $e = \frac{1}{4}BLV$, Q is positive with respect to P

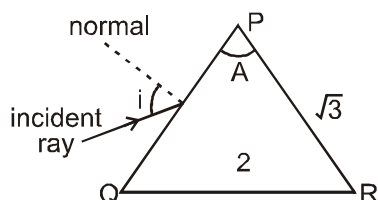
4. In each situation of column-I, an incident wavefront and its corresponding reflected or refracted wavefront is shown. In column-II the optical instrument used for reflection or refraction is given. Always take the optical instrument to the right of incident wavefront. The incident wavefront is moving towards right. Match each pair of incident and reflected/refracted wavefront in column-I with the correct optical instrument given in column-II.

Column-I	Column-II
Incident wavefront	Reflected/Refracted Wavefront
Optical instrument used	



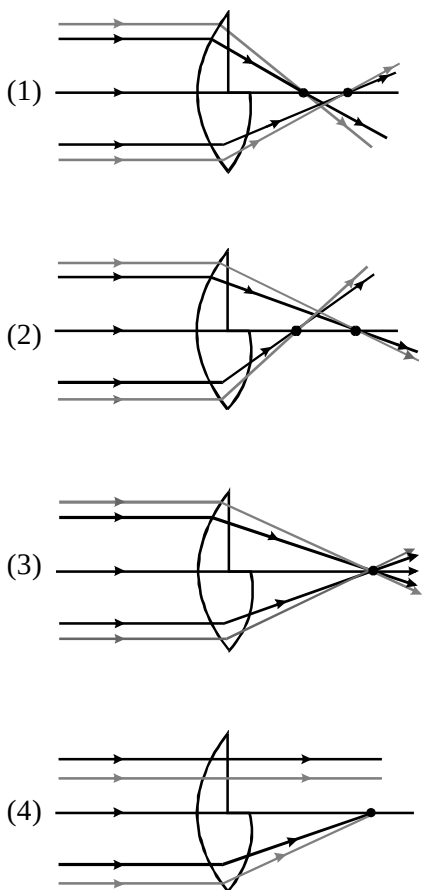
- (1) (A) p, r (B) q, s (C) q, r (D) p, s
 (2) (A) p, r (B) s (C) q, r (D) p
 (3) (A) q, s (B) p, r (C) p, s (D) q, r
 (4) (A) q, s (B) r (C) p, s (D) q

5. A prism of angle A and refractive index 2 is surrounded by medium of refractive index $\sqrt{3}$. A ray is incident on side PQ at an angle of incidence i ($0 \leq i \leq 90^\circ$) as shown. The refracted ray is then incident on side PR of prism. The minimum angle A of prism for which ray incident on side PQ does not emerge out of prism from side PR (for any value of i) is

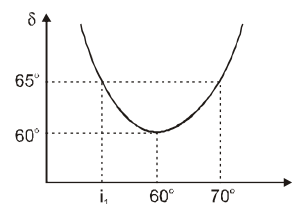


- (1) 30° (2) 45° (3) 60° (4) 120°

6. Choose the correct ray diagram of an equi convex lens which is cut as shown.



7. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Pick up the incorrect statement.

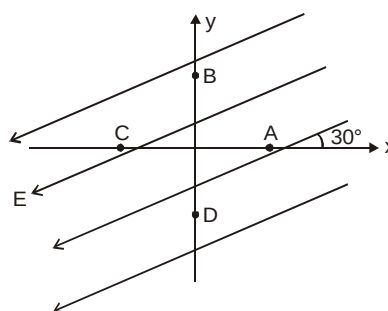


- (1) The angle of prism is 60°
 (2) The refractive index of the prism is $n = \sqrt{3}$
 (3) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
 (4) The curve of ' δ ' vs ' i ' is parabolic

8. There are three concentric thin spheres of radius a, b, c ($a > b > c$). The total surface charge densities on their surfaces are $\sigma, -\sigma, \sigma$ respectively. The magnitude of electric field at r (distance from centre) such that $a > r > b$ is :

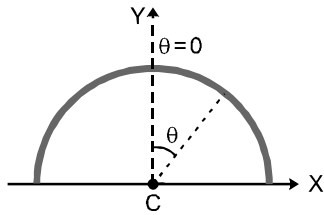
- (1) 0 (2) $\frac{\sigma}{\epsilon_0 r^2} (b^2 - c^2)$
 (3) $\frac{\sigma}{\epsilon_0 r^2} (a^2 + b^2)$ (4) none of these

9. There exists a uniform electric field in the space as shown. Four points A, B, C and D are marked which are equidistant from the origin. If V_A, V_B, V_C and V_D are their potentials respectively, then



- (1) $V_B > V_A > V_C > V_D$ (2) $V_A > V_B > V_D > V_C$
 (3) $V_A = V_B > V_C = V_D$ (4) $V_B > V_C > V_A > V_D$

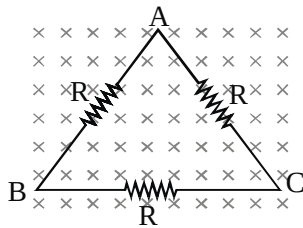
10. A semicircular wire is uniformly charged with linear charge density dependent on the angle θ from y-direction as $\lambda = \lambda_0 |\sin\theta|$, where λ_0 is a constant. The electric field intensity at the centre of the arc is



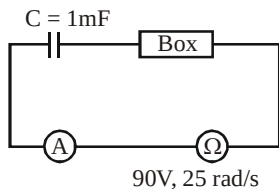
- (1) $\frac{\lambda_0}{2\pi\epsilon_0 R} (+\hat{j})$ (2) $\frac{\lambda_0}{4\pi\epsilon_0 R} (-\hat{j})$
 (3) $\frac{\lambda_0}{6\pi\epsilon_0 R} (-\hat{j})$ (4) $\frac{\lambda_0}{2\sqrt{2}\pi\epsilon_0 R} (-\hat{j})$

11. Three resistances of magnitude R each are connected in the form of an equilateral triangle of side a . The combination is placed in a magnetic field $B = B_0 e^{-\lambda t}$ perpendicular to its plane. The induced current in the circuit is given by

- (1) $\left(\frac{a^2 \lambda}{2\sqrt{3}R} B_0 \right) e^{-\lambda t}$ (2) $\left(\frac{a^2 \lambda}{4\sqrt{3}R} B_0 \right) e^{-\lambda t}$
 (3) $\left(\frac{a^2 B_0}{\lambda 4\sqrt{3}R} \right) e^{-\lambda t}$ (4) $\left(\frac{a^2 B_0 R}{\lambda 4\sqrt{3}} \right) e^{-\lambda t}$

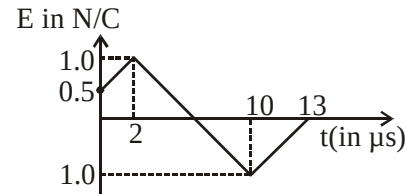


12. In the circuit shown, power factor of circuit is 1 and power factor of box is $\frac{3}{5}$. Find reading of ammeter:-



- (1) 5A (2) 6A (3) 4A (4) 3A

13. The electric field through an area of 2m^2 varies with time as shown in the graph. The greatest displacement current through the area is at



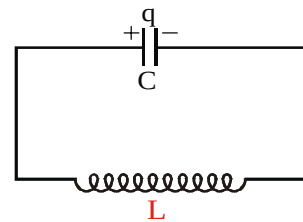
- (1) $t = 1 \text{ sec}$ (2) $t = 4 \text{ sec}$
 (3) $t = 8 \text{ sec}$ (4) $t = 12 \text{ sec}$

14. An inductance L , a capacitance C and a resistance R may be connected to an AC source of angular frequency ω , in three different combination of RC,

RL or RLC in series. Assume that $\omega L = \frac{1}{\omega C}$. The power drawn by the three combinations are P_1 , P_2 & P_3 respectively. Then,

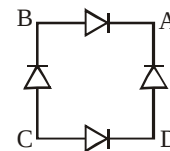
- (1) $P_1 > P_2 > P_3$ (2) $P_1 = P_2 < P_3$
 (3) $P_1 = P_2 > P_3$ (4) $P_1 = P_2 = P_3$

15. In an L-C circuit shown in the figure, $C = 1\text{F}$, $L = 4\text{H}$. At time $t = 0$, charge in the capacitor is 4C and it is decreasing at a rate of $\sqrt{5} \text{ C/s}$. Choose the **CORRECT** statement.



- (1) Maximum charge in the capacitor can be 6C
 (2) Maximum charge in the capacitor can be 8C
 (3) Maximum current in the circuit can be 9 A
 (4) Maximum current in the circuit can be $\frac{2}{3} \text{ A}$

16. For the given circuit shown in fig, to act as full wave rectifier, a.c. input should be connected acrossand.....the d.c. output would appear across.....and.....

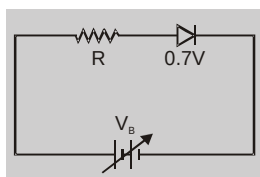


- (1) A, C and B, D (2) B, D and A, C
 (3) A, B and C, D (4) C, A and D, B

17. In a P-N junction diode which is not connected to any circuit-

- (1) Potential is the same every where
- (2) The P-type side is at a higher potential than the N-type side
- (3) There is an electric field at the junction directed from the N-type side to the P-type side
- (4) There is an electric field at the junction directed from the P-type side to the N-type side

18. Assume that the junction diode in the following circuit requires a minimum current of 1 mA to be above the knee point (0.7V) of its I-V characteristic curve. Also assume that the voltage across the diode is independent of current above the knee point. If $V_B = 5V$, what should be the maximum value of R so that the voltage is above the knee point-

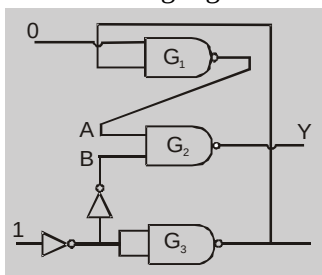


- (1) 4.3 k Ω (2) 860 k Ω (3) 4.3 Ω (4) 860 Ω

19. A N-P-N transistor is connected in common emitter configuration in which collector supply is 8 volt and the voltage drop across the load resistor of 800 ohm connected to the collector circuit is 0.8 volt. If current amplification factor α is $\frac{25}{26}$ and the input resistance of the transistor is 200 ohm then the collector emitter voltage, base current, the voltage and power gain are-

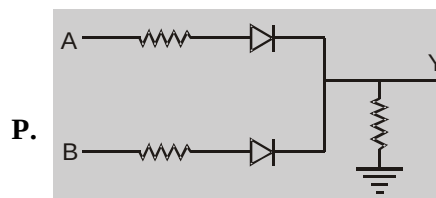
- (1) 3.5 V, 2×10^{-5} A and $A_V = 50$, $A_P = 6500$
- (2) 7.2 V, 4×10^{-5} A and $A_V = 100$, $A_P = 2500$
- (3) 4.5 V, 3×10^{-5} A and $A_V = 50$, $A_P = 6500$
- (4) 5.6 V, 3×10^{-5} A and $A_V = 60$, $A_P = 7500$

20. In circuit in following figure the value of Y is-

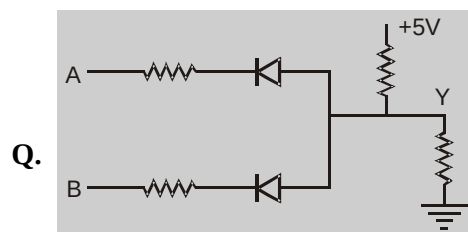


- (1) zero
- (2) 1
- (3) fluctuates between 0 and 1
- (4) indeterminate as the circuit cannot be realized

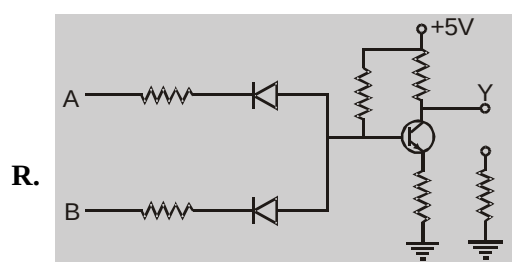
21. Match the operation achieved by the following logic circuit :-



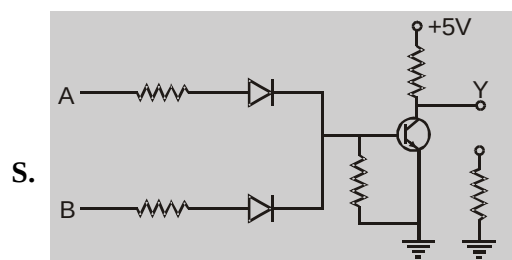
I. AND Gate



II. OR Gate



III. NOR Gate



IV. NAND Gate

- (1) P-II, Q-I, R-IV, S-III
- (2) P-I, Q-II, R-IV, S-III
- (3) P-II, Q-I, R-III, S-IV
- (4) P-I, Q-II, R-III, S-IV

22. A bar magnet of magnetic moment 0.5 A-m² vibrates in an oscillation magnetometer with a period of 2.0 s, under the influence of a uniform magnetic field of induction $\pi \times 10^{-5}$ T. The moment of inertia of the magnet about its vibrational axis is :-

- (1) 0.16×10^{-5} kg-m² (2) 1.6×10^{-5} kg-m²
- (3) 0.62×10^{-5} kg-m² (4) 5.2×10^{-5} kg-m²

23. A nucleus (of nuclear density ρ) disintegrates into two daughter nuclei with masses in the ratio 8 : 27. Density of the smaller nucleus is :-

(1) $\frac{2}{3}\rho$ (2) $\frac{2}{5}\rho$ (3) $\frac{8}{27}\rho$ (4) ρ

24. What is the approximate power output of ${}_{92}\text{U}^{235}$ reactor if it takes 30 days to use up to 2 kg of fuel and if each fission gives 185 MeV of usable energy? Avogadro's number = 6.02×10^{26} per kilomole:-

(1) 45 mega watt (2) 59 mega watt
(3) 72 mega watt (4) 92 mega watt

25. λ_0 is the threshold wavelength of a metal for photoelectron emission. If the metal is irradiated by the light of wavelength λ , of course shorter than λ_0 , the minimum de-Broglie wavelength associated with the ejected electron will be
(c = velocity of light, m = electronic mass)

(1) $\sqrt{\frac{h(\lambda_0 - \lambda)}{2c\lambda\lambda_0}}$ (2) $\sqrt{\frac{2cm(\lambda_0 - \lambda)}{h\lambda\lambda_0}}$

(3) $\sqrt{\frac{h\lambda\lambda_0}{2cm(\lambda_0 - \lambda)}}$ (4) $\sqrt{\frac{h\lambda\lambda_0}{2m(\lambda_0 - \lambda)}}$

26. The ratio of de-Broglie wavelength of molecules of hydrogen and helium moving at rms speed in two gas jars kept separately at temperature 27°C and 127°C , respectively, is :

(1) $\frac{2}{\sqrt{3}}$ (2) 2 : 3 (3) $\frac{\sqrt{3}}{4}$ (4) $\sqrt{\frac{8}{3}}$

27. A certain radioactive sample is observed to undergo 10000 decays in 10 sec. In which of the following cases can we say that decay rate during this time interval is approximately constant and equal to 1000 dps.

(1) $t_{1/2} = 10$ sec (2) $t_{\text{mean}} = 10$ sec
(3) $t_{1/2} \gg 10$ sec (4) $t_{1/2} \ll 10$ sec

28. A galvanometer G deflects full scale when a potential difference of 0.50 V is applied. The internal resistance of the galvanometer r_g is 25 ohms. An ammeter is constructed by incorporating the galvanometer and an additional resistance R_s . The ammeter deflects full scale when a measurement of 2.0 A is made. The resistance R_s is closest to

(1) 0.25 Ω (2) 2.5 Ω
(3) 0.45 Ω (4) 0.1 Ω

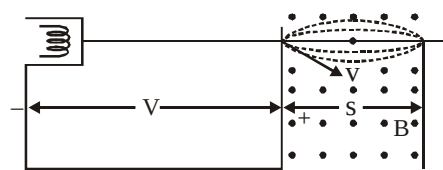
29. In the metre bridge the observations are taken by interchanging the Resistance Box and the unknown resistance so that :-

- (1) The end error (error in reading on the scale) can be removed
(2) The error due to different cross-sectional area can be removed
(3) The error due to resistance of copper-strips / terminal can be removed
(4) All of these

30. A condenser of capacity 6 μF is fully charged using a 6-volt battery. The battery is removed and a resistanceless 0.2 mH inductor is connected across the condenser. The current which is flowing through the inductor when one-third of the total energy is in the magnetic field of the inductor is :-

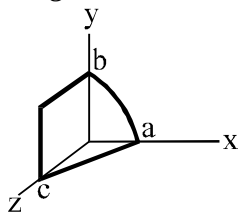
(1) 0.1 A (2) 0.2 A
(3) 0.4 A (4) 0.6 A

31. Figure shows an electron gun that emits electrons at very small angle with axis. The paraxial electrons emitted from the cathode (filament) are refocused at a distance s after accelerating through a potential difference V , s is given by (Take $\pi^2 = 10$) :-



(1) $s = 2\sqrt{\frac{20mV}{eB^2}}$ (2) $s = \sqrt{\frac{20mV}{eB^2}}$
(3) $s = \sqrt{\frac{10mV}{eB^2}}$ (4) None of these

32. In given figure, a wire loop has been bent so that it has three segments: segment ab (a quarter circle), bc (a square corner), and ca (straight). Here are three choices for a magnetic field through the loop:



- (a) $\vec{B}_1 = 3\hat{i} + 7\hat{j} - 5t\hat{k}$ (b) $\vec{B}_2 = 5t\hat{i} - 4\hat{j} - 15\hat{k}$
(c) $\vec{B}_3 = 2\hat{i} - 5t\hat{j} - 12\hat{k}$

where $\vec{B}$ is in milliteslas and t is in seconds. If the induced current in the loop due to $\vec{B}_1$, $\vec{B}_2$ and $\vec{B}_3$ are i_1 , i_2 and i_3 respectively then :-

- (1) $i_1 > i_2 > i_3$ (2) $i_2 > i_1 > i_3$
(3) $i_3 > i_2 > i_1$ (4) $i_1 = i_2 = i_3$
33. In hydrogen atom, electron excites from ground state to higher energy state and its orbital velocity is reduced

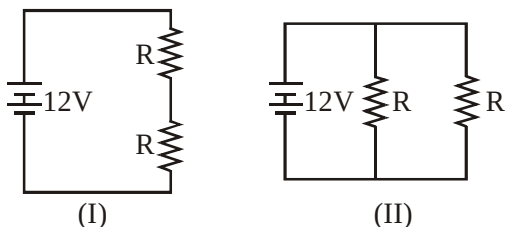
to $\frac{1}{3}$ rd of its initial value. The radius of the orbit in the ground state is R . The radius of the orbit in that higher energy state is :

- (1) $2R$ (2) $3R$ (3) $27R$ (4) $9R$
34. In a p-type semiconductor the acceptor level is situated 57 meV above the valence band. The maximum wavelength of light required to produce a hole will be -

($h = 6.62 \times 10^{-34}$ J-s, $c = 3 \times 10^8$ m/s, $e = 1.6 \times 10^{-19}$ C)

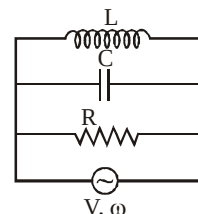
- (1) 57 Å (2) 57×10^{-3} Å
(3) 217763 Å (4) 11.61×10^{-33} m

35. Two identical resistors with resistance R are connected in the two circuits drawn. The battery in both circuits is a 12 volt ideal battery. Which statement is correct?



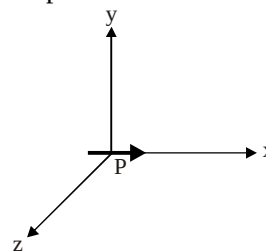
- (1) More current will flow through each R in circuit I than in circuit II.
(2) More total power will be delivered by the battery in circuit II than in circuit I.
(3) The potential drop across each R will be greater in circuit I than in circuit II.
(4) The power dissipated in each R will be greater in circuit I than in circuit II.

36. Consider L, C, R circuit as shown in figure, with a.c. source of peak value V and angular frequency ω . Then the peak value of current through the ac source :-



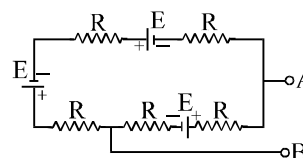
- (1) $\frac{V}{\sqrt{\frac{1}{R^2} + \left(\omega L - \frac{1}{\omega C}\right)^2}}$ (2) $V \sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2}$
(3) $\frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$ (4) $\frac{VR\omega C}{\sqrt{\omega^2 C^2 + R(\omega^2 C^2 - 1)^2}}$

37. A short electric dipole is oriented along x-direction at origin. At which of following point the electric field have no x component.



- (1) (1, 1, 0) (2) $(\sqrt{2}, 1, 0)$
(3) $(1, \sqrt{2}, 0)$ (4) (1, 0, 0)

38. In the circuit shown, all five resistance have the same value R ohms and each cell has an emf E volts. Now AB is short circuited by resistance less wire, find the current in this wire.



- (1) zero (2) $\frac{E}{6R}$ (3) $\frac{E}{16R}$ (4) $\frac{E}{5R}$

39. A nucleus with mass number 220 initially at rest emits an α - particle. If the Q value of reaction is 7.8 MeV and subsequently after the emission of α - particle, a photon of energy 1.2 MeV is also emitted. The kinetic energy of α - particle is ?

(1) 6.48 MeV (2) 7.6 MeV
(3) 0.14 MeV (4) 0.12 MeV

40. Choose the **CORRECT** expression for the ampere-maxwell law.

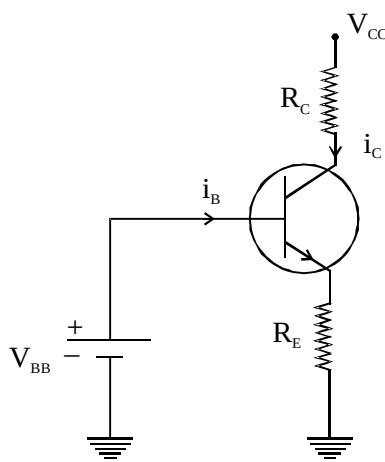
(1) $\oint \vec{E} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt} + \mu_0 i_{\text{enclosed}}$

(2) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_B}{dt} + \mu_0 i_{\text{enclosed}}$

(3) $\oint \vec{E} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt} - \mu_0 i_{\text{enclosed}}$

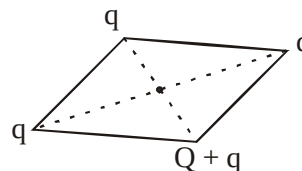
(4) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt} + \mu_0 i_{\text{enclosed}}$

41. For the given bi-polar junction transistor. Find the approximate value of I_B , if $V_{CC} = 9V$, $V_{BB} = 1.5V$, $R_C = 13K\Omega$, $R_E = 17K\Omega$ and $\beta = 100$. (Assume transistor in active mode & $V_{BE} = 0.7V$)



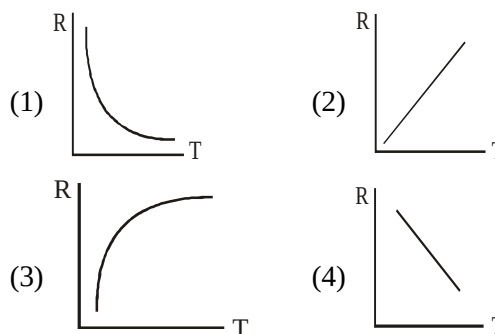
(1) $3 \times 10^{-4} \text{ mA}$ (2) $5 \times 10^{-4} \text{ mA}$
(3) $6 \times 10^{-4} \text{ mA}$ (4) $2 \times 10^{-4} \text{ mA}$

42. For given arrangement, where four charge fixed at ends of a square as given, find value of additional charge Q to be put on one of the vertices so that component of net electric field along the vertical symmetric axis is zero at every point on the vertical symmetric axis.

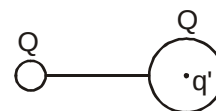


(1) $-q$ (2) $-3q$ (3) $-5q$ (4) $-4q$

43. Assume a hypothetical wire in which free electron density changes with temperature in proportionality $n \propto T$ assuming τ (Relaxation time of collision) and dimensions of wire remain unchanged with increasing temperature. Which one of the resistance v/s temperature graph is true.

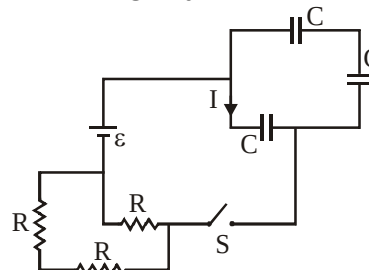


44. Two conducting hollow sphere of radius R and 3R are found to have Q charge on outer surface when both are connected with a long wire and q' charge is kept at the centre of bigger sphere. Then which one is true

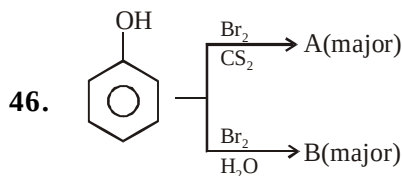


(1) $q' = 2Q$ (2) $q' = 3Q$
(3) $q' = 4Q$ (4) $q' = 6Q$

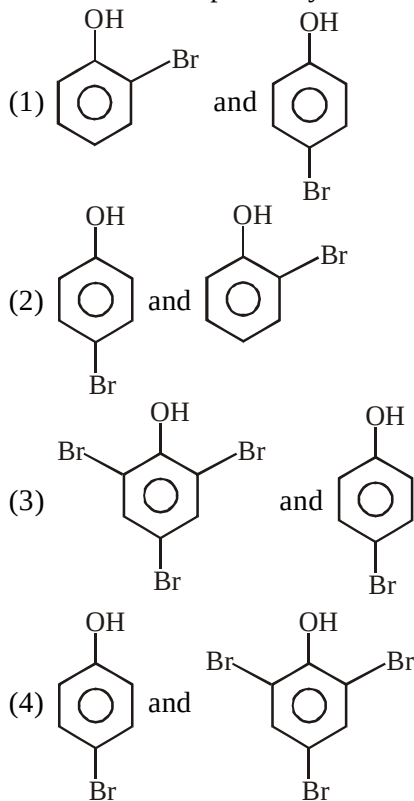
45. In the given circuit switch S is closed at $t = 0$. The current I in the figure just after closing the switch is



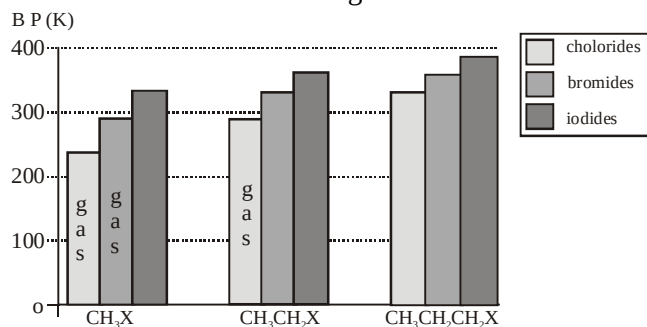
(1) $i = \frac{\epsilon}{2R}$ (2) $i = \frac{3\epsilon}{2R}$
(3) $i = \frac{\epsilon}{2R}$ (4) $i = \frac{\epsilon}{R}$



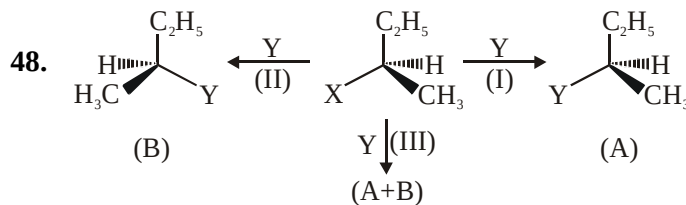
A and B are respectively :



47. Which of the following statements is/are best suited for the following :



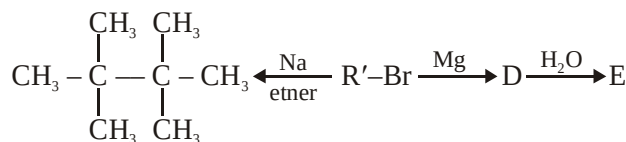
- (1) The attraction gets stronger as molecules get bigger in size.
- (2) The boiling point of alkyl halide is in order of RI > R-Br > R-Cl > R-F
- (3) Magnitude of vander Waal's forces increases as mass of halogen atom increases.
- (4) All of the above



Choose the correct option for I, II and III reaction:

	(I)	(II)	(III)
(1)	Retention of configuration	Inversion of configuration	Racemization
(2)	Retention of configuration	Racemization	Inversion of configuration
(3)	Racemization	Retention of configuration	Inversion of configuration
(4)	Racemization	Inversion of configuration	Retention of configuration

49. Identify D and E in the following reaction :

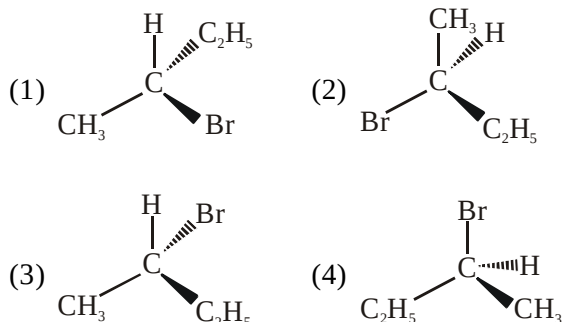
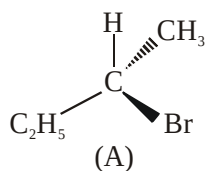


D

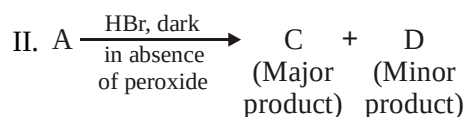
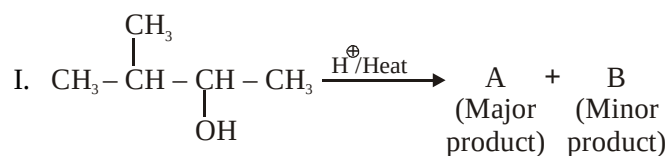
E

- (1) $\text{CH}_3-\text{CH}(\text{MgBr})-\text{CH}_3$ $\text{CH}_3-\text{CH}-\text{H}$
- (2) $\text{CH}_3-\text{CH}(\text{MgBr})-\text{CH}_3$ $\text{CH}_3-\text{CH}-\text{OH}$
- (3) $\text{CH}_3-\text{C}(\text{MgBr})(\text{CH}_3)-\text{CH}_3$ $\text{CH}_3-\text{C}-\text{H}$
- (4) $\text{CH}_3-\text{C}(\text{MgBr})(\text{CH}_3)-\text{CH}_3$ $\text{CH}_3-\text{C}-\text{OH}$

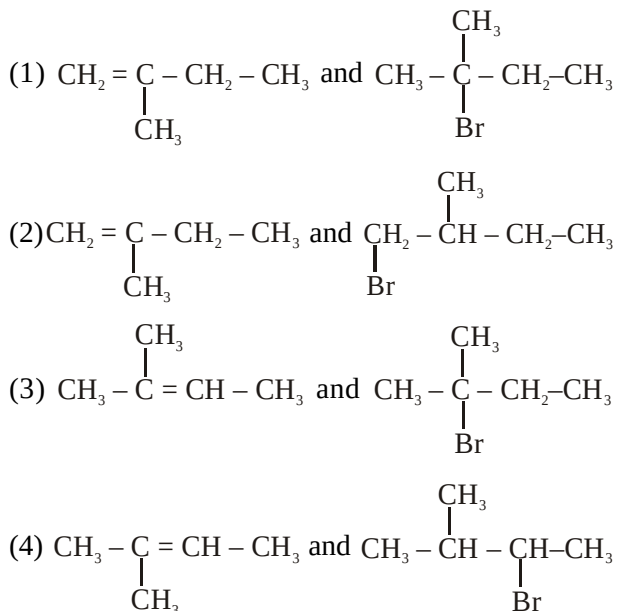
50. Which of the following structures is enantiomeric with the molecule (A) given below ?



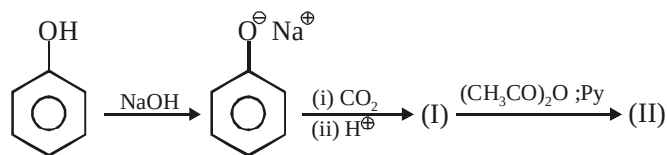
51. In the following reactions :



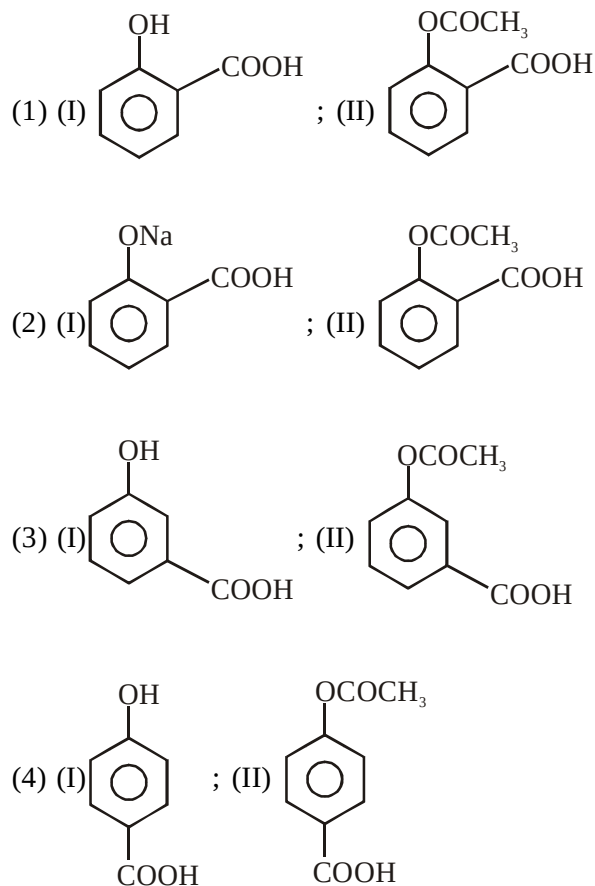
The major products (A) and (C) are respectively:



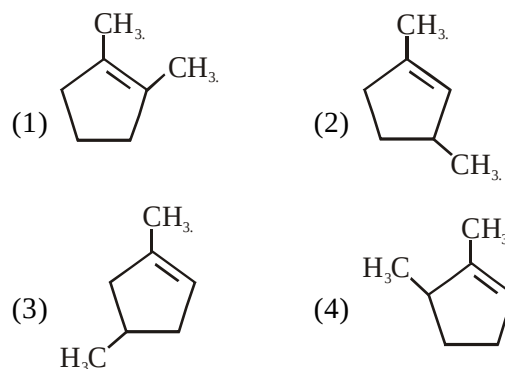
52. In the given reaction :



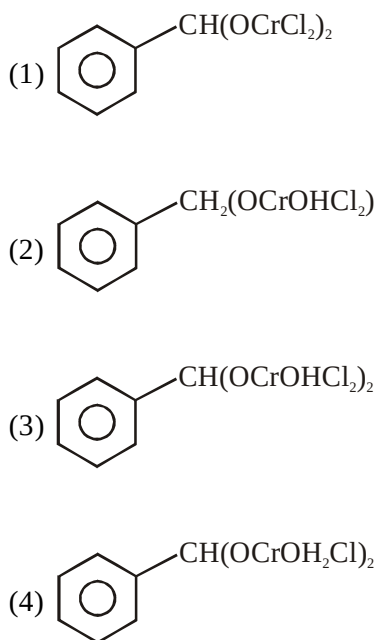
Identify (I) and (II) in the above reaction.



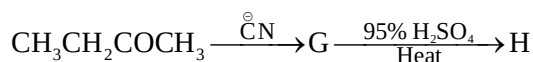
53. Which compound would give 5-keto-2-methyl hexanal upon ozonolysis ?



54. Select the structure of chromium complex formed. When the toluene reacts with chromyl chloride to give benzaldehyde on hydrolysis :

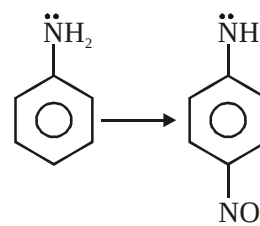


55. The major product H in the given reaction sequence is :

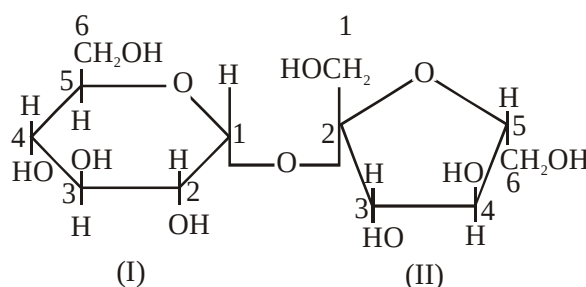


- (1) $\text{CH}_3 - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{COOH}$
- (2) $\text{CH}_3 - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{CN}$
- (3) $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_3}{\overset{\text{OH}}{\text{C}}} - \text{COOH}$
- (4) $\text{CH}_3 - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{CONH}_2$

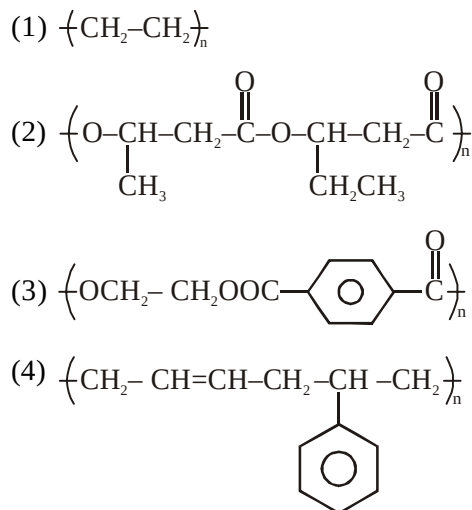
56. Select the correct order of the reagent to carry out the given reaction :



- (1) (i) $(\text{CH}_3\text{CO})_2\text{O}$; Pyridine (ii) $\text{HO}^\ominus$ or $\text{H}^\oplus$ (iii) HNO_3 ; H_2SO_4 ; 288K
- (2) (i) $(\text{CH}_3\text{CO})_2\text{O}$; Pyridine (ii) HNO_3 ; H_2SO_4 ; 288K (iii) $\text{HO}^\ominus$ or $\text{H}^\oplus$
- (3) (i) HNO_3 ; H_2SO_4 ; 288K (ii) $\text{HO}^\ominus$ or $\text{H}^\oplus$
- (4) (i) $\text{HO}^\ominus$ or $\text{H}^\oplus$ (ii) HNO_3 ; H_2SO_4 ; 288K
57. Which one of the following statements is correct about disaccharide ?



- (1) Ring I is furanose with β -glycosidic link
- (2) Ring II is furanose with β -glycosidic link
- (3) Ring I is pyranose with β -glycosidic link
- (4) Ring II is pyranose with β -glycosidic link
58. Which of the following polymers has ethylene glycol as one of the monomer units ?



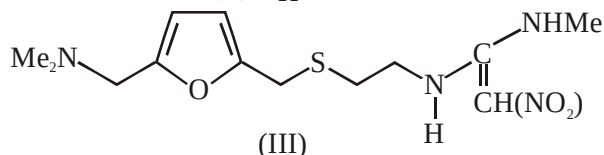
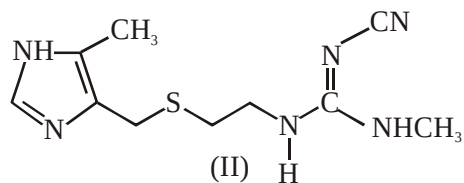
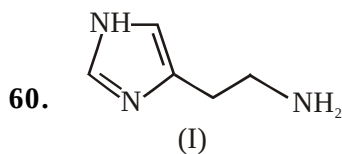
59. Match the polymers given in Column-I with their repeating units given in Column-II. Choose the correct option from the codes given below :

Column-I

Column-II

- (A) Polystyrene (p) $\left(\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CN}) \right)_n$
- (B) Novolac (q) $\left(\text{CH}_2-\text{CH}(\text{C}_6\text{H}_5) \right)_n$
- (C) Buna-N (r) $\left(\text{CH}_2-\text{CH}(\text{CN}) \right)_n$
- (D) Acrilan (s) $\left(\text{C}_6\text{H}_4(\text{OH})-\text{CH}_2-\text{C}_6\text{H}_4(\text{OH})-\text{CH}_2 \right)_n$

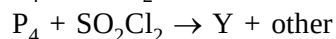
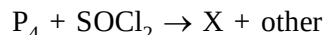
	(A)	(B)	(C)	(D)
(1)	s	r	q	p
(2)	r	p	s	q
(3)	p	q	r	s
(4)	q	s	p	r



In the above structure I, II and III are :

	I	II	III
(1)	Histamine	Cimetidine	Ranitidine
(2)	Histamine	Ranitidine	Cimetidine
(3)	Cimetidine	Ranitidine	Histamine
(4)	Cimetidine	Histamine	Ranitidine

61. In a following chemical reaction



Choose correct option is :-

- (1) X & Y are sp^3 & sp^3d hybridised respectively
 (2) In X & Y central atom is in its maximum oxidation state
 (3) In X & Y all bond lengths are individually similar
 (4) In X & Y, dz^2 orbital participate in the hybridisation
62. Cl_2 can't be prepared by :-
 (1) $\text{HCl} + \text{O}_2 \xrightarrow{\text{CuCl}_2}$
 (2) $\text{NaCl} + \text{F}_2 \longrightarrow$
 (3) $\text{NaCl} + \text{Conc. H}_2\text{SO}_4 \longrightarrow$
 (4) $\text{MnO}_2 + \text{Conc. H}_2\text{SO}_4 + \text{HCl} \longrightarrow$
63. Au & Pt are soluble in aqua regia & forms X & Y complex respectively. Hybridisation in X & Y respectively.
 (1) sp^3 & sp^3 (2) dsp^2 & sp^3d^2
 (3) d^2sp^3 & d^2sp^3 (4) dsp^2 & d^2sp^3
64. Which of the following elements have least difference in E^0 value of M^{+2}/M :-
 (1) V & Mn (2) Mn & Fe
 (3) Fe & Co (4) Co & Ni
65. Incorrect about element 'Gd' is [Atomic No = 64]
 (1) It has a maximum no. of unpaired e^- in 4f series
 (2) It is a paramagnetic as well as ferro-magnetic in nature
 (3) It is only synthetic radio-active element in 4f series
 (4) Its +3 oxidation state is stable
66. Which of the following pair of species have same hybridisation ?
 (1) $[\text{Ni}(\text{CO})_4]$ & $[\text{Co}(\text{CO})_4]$
 (2) $[\text{Ni}(\text{NH}_3)_4]^{2+}$ & $[\text{Rh}(\text{PPh}_3)_3\text{Cl}]$
 (3) $[\text{NiCl}_4]^{2-}$ & $[\text{Pt}(\text{CN})_4]^{2-}$
 (4) All of these
67. Which of the following on hydrolysis gives HOCl ?
 (1) NCl_3 (2) PCl_3 (3) BCl_3 (4) BiCl_3

68. Which of the following does not decolorise Iodine?
 (1) NaBr (2) NaOH
 (3) $\text{Na}_2\text{S}_2\text{O}_3$ (4) All of these
69. Considering Ellingham diagram, which of the following metal can be used to reduce alumina ?
 (1) Fe (2) Zn (3) Mg (4) Cu
70. Incorrect option regarding 3D-silicate is :-
 (1) Number of oxygen shared per tetrahedral unit is 4.
 (2) It forms 6 membered ring
 (3) Covalency of central atom is 4
 (4) Zeolite is an example of 3D-silicate
71. If a metal has low affinity towards oxygen then purification of metal may be carried out by :-
 (1) Bessemerisation (2) Distillation
 (3) Cupellation (4) Poling
72. Which of the following inert gas does'nt form clathrate compound ?
 (1) [Xe] (2) [Ar] (3) [Kr] (4) [He]
73. O_2 is formed in the hydrolysis of :-
 (1) XeF_2 (2) XeF_4 (3) F_2 (4) All of these
74. When H_2S is passed through nitric acid solution, the product formed is :-
 (1) SO_3 (2) Colloidal sulphur
 (3) SO_2 (4) Rhombic sulphur
75. Find the name of a member of the lanthanide series which is well known to exhibit +4 oxidation state :-
 (1) Ce (2) Th (3) Eu (4) Yb
76. A few substances are attracted very strongly by a magnetic field called ferromagnetic substances. Which of the following statement is incorrect for ferromagnetic substances :-
 (1) In an unmagnetised piece of a ferromagnetic substance the domains are randomly oriented and their magnetic moments get cancelled
 (2) Each domain act as a tiny magnet
 (3) When the substance is placed in a magnetic field all the domains get oriented in the opposite direction of the magnetic field
 (4) Examples of ferromagnetic substances are iron, cobalt, nickel, gadolinium & CrO_2
77. Gold (atomic radius = 0.144 nm) crystallises in a face-centered unit cell. What is the length of side of the cell ?
 (1) $2\sqrt{2} \times 0.144 \text{ nm}$ (2) $\frac{4 \times 0.144 \text{ nm}}{\sqrt{2}}$
 (3) $\sqrt{2} \times 0.288 \text{ nm}$ (4) All of the above
78. Which of the following statement is incorrect for electrolysis of sulphuric acid using platinum electrode:-
 (1) In case of dilute sulphuric acid oxygen gas evolved at anode
 (2) In case of higher concentration of sulphuric acid $\text{S}_2\text{O}_8^{2-}$ is obtained at anode
 (3) In case of dilute sulphuric acid hydrogen gas evolved at cathode
 (4) In case of concentrated sulphuric acid sulphur is obtained at cathode
79. Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane methanol etc. directly into electrical energy are called:-
 (1) Dry cell
 (2) Electrolytic cell
 (3) Fuel cell
 (4) Mercury cell
80. If a gas like O_2 , H_2 , CO , Cl_2 , NH_3 or SO_2 is taken in a closed vessel containing powdered charcoal, it is observed that the pressure of the gas in the enclosed vessel decreases. It is mainly due to :-
 (1) Absorption
 (2) Adsorption
 (3) Chemical reaction
 (4) Desorption
81. Which of the following is not an application of adsorption :-
 (1) Separation of inert gases
 (2) In curing diseases
 (3) Chromatographic analysis
 (4) Ultrafiltration

82. Among the following select the incorrect statement regarding colloids :-

- (1) Emulsion of oil in water are unstable and sometimes they separate into two layers on standing. For stabilisation emulsifying agent is added
- (2) There are two factors which are responsible for the stability of lyophilic sols
- (3) When excess of an electrolyte is added, the colloidal particles are precipitated
- (4) The process of settling of colloidal particles is called peptization

83. Correct statement regarding catalysis is :-

- (1) The modern adsorption theory of catalysis is the combination of intermediate compound formation theory and the old adsorption theory
- (2) The modern adsorption theory of catalysis does not explain the action of catalytic promoters and catalytic poisons
- (3) Modern adsorption theory of catalysis explains why the catalyst remains unchanged in mass and chemical composition at the end of the reaction
- (4) All of these

84. For a homogeneous gaseous reaction $A \longrightarrow 3B$, if pressure after time t was P_T and after completion of reaction, pressure was P_∞ then select correct relation :-

- (1) $k = \frac{1}{t} \ln \left(\frac{P_\infty}{3(P_\infty - P_T)} \right)$
- (2) $k = \frac{1}{t} \ln \left(\frac{2P_\infty}{(3P_\infty - P_T)} \right)$
- (3) $k = \frac{1}{t} \ln \left(\frac{3P_\infty}{3P_\infty - P_T} \right)$
- (4) $k = \frac{1}{t} \ln \left(\frac{2P_\infty}{3(P_\infty - P_T)} \right)$

85. A reaction takes place in various steps. The rate constant for first, second, third and fourth steps are k_1 , k_2 , k_3 and k_4 respectively. The overall rate constant is given by :-

$$k = \frac{k_2}{k_3} \left(\frac{k_1}{k_4} \right)^{1/2}$$

if activation energy are 40, 60, 50 and 10 kJ/mol respectively, the overall energy of activation (kJ/mol) is :

- (1) 10
- (2) 20
- (3) 25
- (4) None of these

86. When molten lithium chloride (LiCl) is electrolyzed, lithium metal is formed at the cathode. If current efficiency is 75% then how many grams of lithium are liberated when 1930 C of charge pass through the cell ?

(Atomic weight : Li = 7) :-

- (1) 0.105
- (2) 0.120
- (3) 0.28
- (4) 0.240

87. The composition of a sample of Wustite is $Fe_{0.90}O$. What is the percentage of iron present as Fe^{3+} in total iron ?

- (1) 22.22%
- (2) 15%
- (3) 35%
- (4) 45%

88. On mixing 30 mL CCl_4 and 20 mL $CHCl_3$, the total volume of the solution is :-

- (1) > 50 mL
- (2) < 50 mL
- (3) = 50 mL
- (4) Can't be predicted

89. For a reaction $A + 2B \longrightarrow 6C + 2D$, if rate of disappearance of A is $2.6 \times 10^{-2} \text{ M min}^{-1}$. Then

what will be value of $\frac{-\Delta[B]}{\Delta t}$?

- (1) $5.2 \times 10^{-2} \text{ M s}^{-1}$
- (2) $1.3 \times 10^{-2} \text{ M s}^{-1}$
- (3) $5.2 \times 10^{-2} \text{ M min}^{-1}$
- (4) $6.5 \times 10^{-3} \text{ M min}^{-1}$

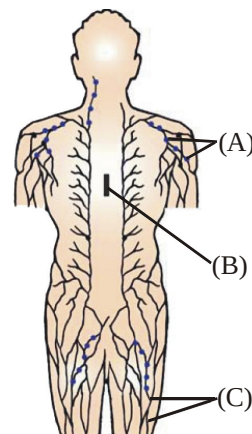
90. Phenol associates in benzene to a certain extent in dimerisation reaction. A solution containing 0.02 kg of phenol in 1.0 kg of benzene has its freezing point depressed 0.69 K. Hence, degree of association of phenol dimerized will be :-

$[K_f (C_6H_6) = 5.12 \text{ K kg mol}^{-1}]$

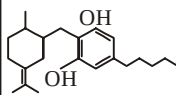
- (1) 0.63
- (2) 0.73
- (3) 0.83
- (4) 0.93

91. Every antibody is specific because?
- Specific type of bonds found in each antibody
 - There are only five type of antigens stimulates to form only 5 types of antibodies (IgG, A, M, D, E)
 - F_c segment is different for each antibody
 - Parotope is specifically designed for each type of epitope in every antibody.
92. The killing of target cells by T-killer cell is?
- Ist line of defence
 - IInd line of defence
 - IIIrd line of defence
 - Agglutination
93. Natural killer cells having granules filled with?
- Pore forming cytolytic protein
 - Lymphotoxins
 - Histamine
 - Cytokines
94. "CLONAL SELECTION" refers to?
- Formation of specific phagocytic cells.
 - Formation of Non-specific clones of T-helper cells.
 - Formation of non-specific clones of T-killer cells.
 - Formation of specific active cells of cell mediated and humoral immunity.
95. Potential disadvantage of using horse immunoglobulins during passive immunity is/are?
- Serum sickness
 - Phagocytic cells found in antiserum also affects cells of human body.
 - Both (1) and (2)
 - It develops autoimmunity.
96. Which of the following humour shows "Hot personality" in person according to good humour hypothesis of Hippocrates?
- Yellow bile
 - Black bile
 - Blood
 - Phelgm

97. Vaccination is :-
- Artificially injected inactivated pathogen to our body.
 - Active immunization
 - Passive immunization
 - Both (1) and (2)
98. In given diagramme identify (A), (B) and (C)



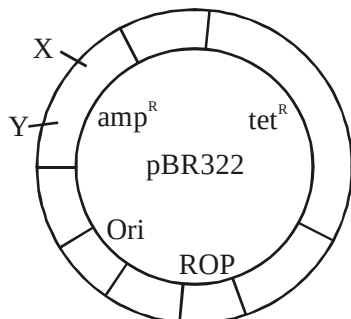
- (A) – Blood vessels, (B) – Tonsils, (C) – Lymph nodes
 - (A) – Lymphatic vessels, (B) – Bone marrow, (C) – Thymus
 - (A) – Lymph nodes, (B) – Thymus, (C) – Lymphatic vessels
 - (A) – Payer's patches, (B) – Spleen, (C) – Blood vessels
99. Which of the following is incorrect matching:-

	Drug	Source	Function
(1)	Nicotine	Tobacco	Stimulates Adrenal gland to release adrenaline
(2)	Ganja	Inflorescence of cannabis	Hallucination
(3)		Latex of <i>Papaver somniferum</i>	Sedative and pain killer
(4)	Cocaine	<i>Erythroxylum</i>	Interferes with the transport of dopamine

100. Which of the following is not related to alcoholism:-

- (i) Amnesia (ii) Gastritis
(iii) Lung Cancer (iv) Liver Cirrhosis
(v) Fatty liver syndrome (vi) Bronchitis
(1) iii, iv, v, vi (2) i, ii, iv, v, vi
(3) iii, vi (4) i, ii, iii, iv, v, vi

101. Identify the restriction enzyme used to cut the site 'X', 'Y' respectively :



- (1) Pvu I, Pst I (2) Sal I, E.CoRI
(3) Pvu II, Pst I (4) BamH I, Sal I

102. Which is not true about PCR (Polymerase chain reaction)?

- (1) PCR can amplify very small amount of DNA
(2) It can be used to detect HIV in suspects
(3) Only RNA can be used as primer
(4) Used to detect mutation in genes

103. In agarose gel electrophoresis :

- (1) DNA migrates towards negative electrode.
(2) Larger molecules migrate faster than smaller molecules
(3) Ethidium bromide can be used to visualize the DNA fragments.
(4) Pink coloured bands of DNA can be visualize when gel exposed to UV light

104. 'Pomato' is developed by which tissue culture method :-

- (1) Meristem culture
(2) Embryo culture
(3) Anther culture
(4) Somatic hybridisation

105. Biogas is the mixture of gases produced by the microbial activity. The type of the gas produced depends upon

- (1) Type of microbes
(2) Type of organic substrate/waste
(3) Size of digester
(4) (1) and (2) both

106. Match the following :

Column I		Column II	
(A)	<i>Aspergillus niger</i>	(i)	Acetic acid
(B)	<i>Acetobacter aceti</i>	(ii)	Butyric acid
(C)	<i>Clostridium butylicum</i>	(iii)	Lactic acid
(D)	<i>Lactobacillus</i>	(iv)	Citric acid

- (1) A-(iii), B-(i), C-(ii), D-(iv)
(2) A-(iv), B-(i), C-(ii), D-(iii)
(3) A-(iv), B-(ii), C-(i), D-(iii)
(4) A-(i), B-(ii), C-(iii), D-(iv)

107. Match the following :

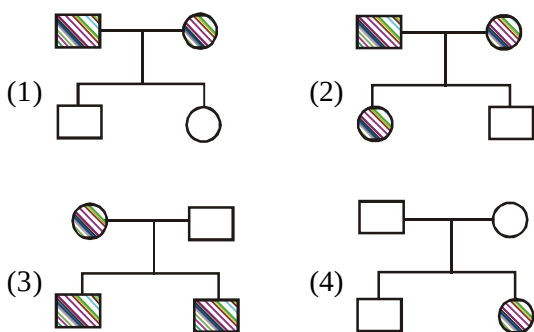
- (A) Totipotency (i) Virus free culture
(B) Explant (ii) Capacity of generate a whole plant
(C) Single cell protein (S.C.P.) (iii) Methylophilus
(D) Meristem culture (iv) Part of plant use in tissue culture

- (1) A-i, B-ii, C-iii, D-iv (2) A-ii, B-iii, C-iv, D-i
(3) A-ii, B-iv, C-iii, D-i (4) A-iii, B-ii, C-iv, D-i

108. Select correct statement :

- (a) *Spirulina* can be grow easily on materials like waste water from potato processing plants
(b) In 2000 maize hybrids that had twice the amount of the amino acids lysine and tryptophan
(c) Mung bean, resistance to yellow mosaic virus, induced by tissue culture
(d) Parbhani Kranti is a variety of Wheat
(1) a and b only (2) a and d only
(3) a and c only (4) All correct except b

109. Lactose composed of :
 (1) Glucose + Galactose (2) Glucose + Fructose
 (3) Glucose + Glucose (4) Glucose + Mannose
110. The most abundant intracellular cation is ;
 (1) K^+ (2) Na^+ (3) Ca^{++} (4) H^+
111. In a random mating population of 2000 individuals, 1280 individual belongs to RR genotype, 640 belongs to Rr genotype while the remaining 80 individual belongs to rr genotype. What will be the frequency of recessive allele ?
 (1) 0.4 (2) 0.6
 (3) 0.8 (4) 0.2
112. The characters which can express in homozygous and heterozygous condition in case of garden pea plant :-
 (a) Axial flower (b) Green seed colour
 (c) Green pod colour (d) Violet flower
 (1) b, c, d (2) a, b, d (3) a, c, d (4) a, b, c, d
113. Source of non-parental phenotypes in Mendelian dihybrid cross is :-
 (1) Dominance
 (2) Mutation
 (3) Independent assortment
 (4) Crossing over
114. Which pedigree is not possible for autosomal dominant disease ?



115. Sickle cell anemia is an example of all except :-
 (1) Autosomal recessive trait
 (2) Co-dominance
 (3) Single base substitution
 (4) Chromosomal mutation

116. If a trihybrid mulatto man marries a white woman, then what will be the phenotype ratio in the offsprings ?
 (1) 1 : 3 : 3 : 1
 (2) 1 : 2 : 1 : 2
 (3) 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
 (4) 1 : 6 : 15 : 20 : 15 : 6 : 1
117. When red flowered plants of snapdragon are crossed with pink flowered plants, then ratio of progenies obtained will be :-
 (1) 1 red : 1 white
 (2) 1 red : 1 pink
 (3) All pink
 (4) 1 red : 2 pink : 1 white
118. If there are 6 multiple alleles of a gene then how many types of homozygous conditions are possible in the population ?
 (1) 21 (2) 12
 (3) 6 (4) cannot be predicted
119. Linkage maps are based on frequency of :-
 (1) Non-disjunction (2) Mutation
 (3) Translocation (4) Crossing over
120. How many meiotic divisions are required for the formation of 200 male gametes in capsella (Angiosperm)?
 (1) 25 (2) 50 (3) 100 (4) 200
121. Consider the following four statements (a-d) and select the option which includes all the correct ones only.
 (a) Embryo-sac is also known as megasporophyte.
 (b) After fertilization inner integument of ovule is converted into the intine.
 (c) Unitegmatic ovules are found in gymnosperms.
 (d) Filiform apparatus guides the entry of pollen tube into the synergid.
- Options :-**
 (1) Statements (b), (c) and (d)
 (2) Statements (a), (b)
 (3) Statements (c), (d)
 (4) Statements (a), (c) and (d)

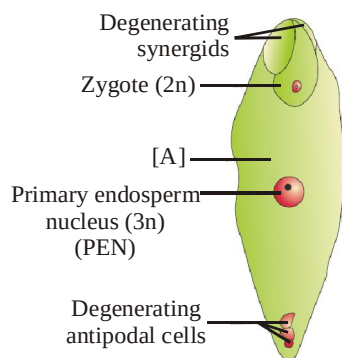
122. The synergids have special cellular thickenings at the micropylar tip called :-

- (1) Egg apparatus (2) Fibres of α -cellulose
(3) Filiform apparatus (4) Obturators

123. Male and female gametes in flowering plants are produced in the :-

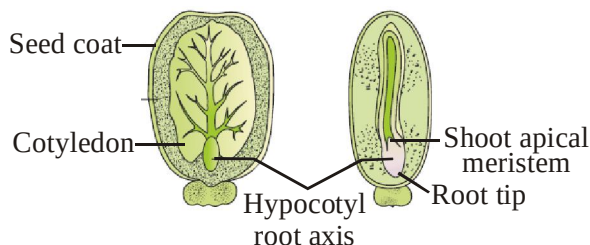
- (1) Ovule and microsporangium respectively
(2) Pollen grain and embryo sac respectively
(3) Microsporangium and anther respectively
(4) Embryo sac and male gametophyte respectively.

124. Following diagram represents fertilized embryo sac. Identify labelled structure [A] of this diagram:-



- (1) Endosperm
(2) Perisperm
(3) Primary endosperm cell
(4) Central cell

125. Following figure represents :-



- (1) Endospermic seed of bean
(2) Endospermic seed of castor
(3) Non-endospermic seed of bean
(4) Non-endospermic seed of castor

126. Ruminant endosperm occurs in the seed of :-

- (1) Areaceae (2) Poaceae
(3) Myrtaceae (4) Euphorbiaceae

127. In angiosperm development of female gametophyte occurs from :-

- (1) Microspore (2) Megaspore
(3) Endospore (4) Exospore

128. How many meiotic divisions are required for the formation of 100 megaspores in *Capsella* ?

- (1) 25 (2) 50 (3) 100 (4) 200

129. Murrah and Nagpuri are breeds of –

- (1) Buffalo (2) Cattle (3) Goat (4) Horse

130. Blue revolution is concerned with –

- (1) Fish culture (2) Apiculture
(3) Sericulture (4) Moriculture

131. Which of the following product is not obtained from bee keeping?

- (1) Honey (2) Wax
(3) Propolis and wax (4) Lac and resin

132. Placenta is the region where :-

- (1) Foetus is attached to myometrium of uterus
(2) Foetus receives oxygen and nourishment from mother's blood
(3) Foetus is covered by fetal membranes
(4) Foetus is attached to mother by spermatocord

133. Non-identical twins will be produced when :-

- (1) Two eggs are fertilized
(2) One ovum is fertilized by two sperms
(3) One sperm fertilizes two ova
(4) One fertilized egg divided into two parts

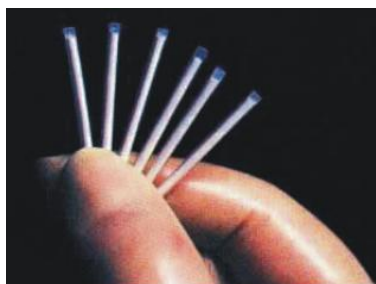
134. Degeneration of corpus luteum occurs due to inhibition of secretion of hormone :-

- (1) FSH (2) LH
(3) Estrogen (4) Progesterone

135. Amniocentesis can diagnose which of the following defects :-

- (a) Down syndrome (b) Metabolic disorder
(c) Blue baby syndrome (d) Sex-determination
(1) a, b and c (2) b, c and d
(3) only a and c (4) a, b and d

136. Identify the contraceptive device shown below as well as the related right place of its implantation into a woman and select the correct option for the two together :



	Contraceptive device	Site of implant
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Implants	Subcutaneous
(4)	Multiload 375	Uterine wall

137. Which of the following contributes in the formation of seminal plasma ?

- (a) Sertoli cells (b) Seminal vesicle
(c) Spermatogonia (d) Leydig cells
(e) Bulbourethral gland
(f) Prostate gland

- (1) b, c, e, f (2) a, b, c, f
(3) b, c, d, e, f (4) only b, e, f

138. The key events in embryo development are given below. Which is the correct order of sequences?

- (i) Organogenesis (ii) Fertilization
(iii) Blastulation (iv) Gastrulation
(v) Cleavage

- (1) v → ii → iv → i → iii
(2) ii → iii → v → i → iv
(3) iii → iv → ii → i → v
(4) ii → v → iii → iv → i

139. What will be happened, if we remove zona pellucida of a embryo in morula stage ?

- (1) Embryo may be destroyed in fallopian tube
(2) Embryo may be implanted at abnormal site
(3) Implantation will not be possible
(4) Both (1) and (2)

140. Read carefully the statements given below for test tube baby and answer the given question.

- (A) Fertilization inside female genital tract and growth in test tube.
(B) Gestation period is same as a normal
(C) Fertilization outside and gestation inside uterus of mother.
(D) Both fertilization and development are outside the female genital tract.

How many statements are **correct** ?

- (1) 2 (2) 3
(3) 4 (4) 1

141. The immune system does not normally attack on immature male germ cells, because :-

- (1) They are recognized as "self" structures
(2) They do not have any antigens on their cell membrane
(3) These cells are protected by the blood testis barrier
(4) The acrosome covers many antigens that would be recognized as foreign

142. Identify the wrongly matched pair :

- (1) i gene of lac operon - Non-constitutive gene
(2) untranslated region - at 3' and 5' end of mRNA in eukaryotes
(3) 28S rRNA of eukaryotes - Catalytic role
(4) AUG - Non degenerate codon

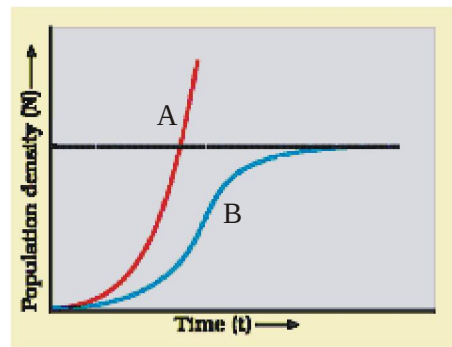
143. Unique property of the DNA polymerase-I is :

- (1) 5' → 3' polymerase activity
(2) 3' → 5' polymerase activity
(3) 5' → 3' exonuclease activity
(4) 3' → 5' exonuclease activity

144. During translation which step do not require GTP :

- (1) Activation of Amino acid
(2) Releasing of polypeptide chain
(3) Translocation
(4) Binding of charged tRNA with mRNA

- 145.** Nitrogen base of codon that is specific in anticodon recognition is/are :
 (1) First nitrogen base
 (2) Second nitrogen base
 (3) Third nitrogen base
 (4) 1 & 2
- 146.** Pyrimidine that is common between DNA & RNA:
 (1) A (2) G (3) C (4) U
- 147.** Sperm of human have approx.....nucleosome:
 (1) 3.3×10^7 (2) 6.6×10^7
 (3) 1.65×10^9 (4) 1.65×10^7
- 148.** RNA-Polymerase is required for the how many given process:
 (A) Replication (B) Transcription
 (C) Translation (D) DNA finger printing
 (1) Two (2) Three
 (3) Four (4) One
- 149.** Out of the total number of gene estimated in human genome, nearly 10% are located on :
 (1) Chromosome 1 (2) Chromosome X
 (3) Chromosome 21 (4) Chromosome Y
- 150.** The probe used initially by Alac Jaffrey during development of DNA finger printing was :
 (1) VNTR (2) rDNA
 (3) Ribozyme (4) SAT chromosome
- 151.** Which trophic level can not be held by human beings in a food chain ?
 (1) First Trophic level (2) Second Trophic level
 (3) Third Trophic level (4) Fourth Trophic level
- 152.** Consider the following statements (A-D) :-
 (A) Ecosystem are exempt from the Second Law of Thermodynamics
 (B) During succession, establishment of new biotic community mainly depends on climate.
 (C) Pyramids of biomass in sea generally inverted.
 (D) In an aquatic ecosystem, DFC is the major conduit for energy flow.
 How many of the above statement(s) is/are false?
 (1) One (2) Two
 (3) Three (4) Four
- 153.** Tropical rain forest, Temperate deciduous forest, Tropical deciduous forest. Tundra, Taiga, Savannah, Sea Coast, Desert; from above given Biomes. How many major latitudinal biomes found in india :-
 (1) 6 (2) 5 (3) 4 (4) 3
- 154.** The process in which detritus degraded into simple inorganic substances by bacterial and fungal enzymes is called _____ :-
 (1) Catabolism (2) Humification
 (3) Fragmentation (4) Leaching
- 155.** Mainly which nutrient cycle affected by human activities ?
 (1) Carbon cycle (2) Phosphorus cycle
 (3) Nitrogen cycle (4) Sulphur cycle
- 156.** Find out odd one interaction :-
 (1) Fig-Wasp
 (2) Egret-Cattle
 (3) Orchid-Mango
 (4) Sea-anemone-Clown fish
- 157.** Identify the A and B in given population growth curve :-



- (1) A growth curve-when resources are limiting
 (2) B growth curve-when resources are not limiting
 (3) A growth curve-when resources are not limiting
 (4) B is exponential growth curve when resource are limiting

- 158.** Which of the following lack physiological ability to deal with high temperature of their habitat ?
 (1) Most mammals
 (2) Most birds
 (3) Desert lizards
 (4) All of these
- 159.** Which of the following organisms breed only once in their life time ?
 (1) Bamboo (2) Salmon fish
 (3) Most birds (4) Both (1) and (2)
- 160.** Choose the correct equation :-
 (1) $NPP = GPP - R$
 (2) $GPP = NPP + R$
 (3) Both (1) and (2)
 (4) $NPP = GPP + R$
- 161.** Amazonian rainforests is being cut and cleared by cultivation of which crop?
 (1) Ground nut (2) Soyabeans
 (3) Maize (4) Rye
- 162.** Which of the followings is a reason for species extinction but not included in evil quarter?
 (1) Habitat loss and fragmentation
 (2) Over exploitation
 (3) Pollution
 (4) Co-extinction
- 163.** Select the incorrect statement from followings.
 (1) Biodiversity hot spots have very high levels of species richness and high degree of endemism.
 (2) In Meghalaya, the sacred groves are the last refuges for a large number of rare and threatened species.
 (3) The convention on biological diversity held in Rio-de-janeiro in 2002.
 (4) In rivet popper hypothesis of paul Ehrlich the airplane represents ecosystem.
- 164.** Select the incorrect match from following?
 (1) National environment (Protection) Act - 1986
 (2) Water (Prevention and control of pollution) Act - 1974
 (3) Montreal protocol effective from - 1987
 (4) Conference of parties - 25 (COP - 25) - Madrid, Spain
- 165.** A process in which the fitness of one species is significantly lower in the presence of another species it is called as:
 (1) Parasitism
 (2) Amensalism
 (3) Competitive release
 (4) Competition
- 166.** A stable community must be resistant to invasion by alien species, In which community establishment of alien species is easily possible:
 (1) Island
 (2) Grassland
 (3) Rain forest
 (4) Tundra
- 167.** Which of the following statement is not true regarding David Tilman's long term ecosystem experiments?
 (1) Plots with more species showed less year to year variation in total biomass
 (2) Increased biodiversity contribute to higher productivity.
 (3) Temporal stability of the ecosystem increased with biodiversity.
 (4) There is no effect of biodiversity on productivity and stability of an ecosystem.

- 168.** Which of the following statement is not true with respect to biodiversity?
- (1) Amphibian's diversity in western ghaat is more than the amphibian diversity in eastern ghaat.
 - (2) Species diversity of insects is more than species diversity of vertebrates
 - (3) India has 2.4% of the world's land area and 8.1% of global species diversity
 - (4) Scandinavian countries have more ecosystem diversity than tropical countries
- 169.** How many of the following are secondary pollutants?
- Ozone, Peroxy acetyl nitrate, Acid rain, CO₂, SO₂, Lead
- (1) Two
 - (2) One
 - (3) Four
 - (4) Three
- 170.** Which of the following is most efficient in removing particulate matter from plume?
- (1) Catalytical converter
 - (2) Electrostatic precipitator
 - (3) Cyclone precipitator
 - (4) Scrubber
- 171.** India is situated in which realm ?
- (1) Oriental
 - (2) Nearctic
 - (3) Ethiopian
 - (4) Neotropical
- 172.** Homologous organs show which type of evolution:-
- (1) Convergent
 - (2) Divergent
 - (3) Parallel
 - (4) None
- 173.** In human-evolution the first man who started agriculture was:-
- (1) Cromagnon man
 - (2) Neanderthal man
 - (3) *Homo sapiens sapiens*
 - (4) Peking man
- 174.** Theory of pangenesis was given by :-
- (1) Lamarck
 - (2) Huxley
 - (3) Darwin
 - (4) Weismann
- 175.** Evolutionary history of a species is also known as:-
- (1) Ontogeny
 - (2) Phylogeny
 - (3) Neotany
 - (4) Mutation
- 176.** Which statement is correct :-
- (1) Lamarck theory-struggle for existence
 - (2) Darwin theory-Use and disuse of organ
 - (3) Biogenetic law-Recapitulation theory
 - (4) Lamarck theory-Theory of continuity of germplasm
- 177.** Which period is also known as "Golden age of fishes"?
- (1) Silurian
 - (2) Ordovician
 - (3) Cambrian
 - (4) Devonian
- 178.** In replica-plate experiment some bacteria were able to grow in penicillin containing medium due to :-
- (1) Natural selection
 - (2) Reproductive isolation
 - (3) Genetic drift
 - (4) Crossing over

179. The first living organism was :-

- (1) Aerobic-heterotrophic
- (2) Aerobic-Autotrophic
- (3) Anaerobic-heterotrophic
- (4) Anaerobic-Autotrophic

180. Near about the carboniferous period all the present day continents were in form of a single large land mass called :-

- (1) Race
- (2) Realm
- (3) Pangaea
- (4) Country



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